Material property based analysis of Human Body Communication in Body Resonance Regime

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***Abstract*—Optimizing the battery life for the miniaturized wearable devices, residing around the human body has been one of the cardinal aspects that drive the research for an energy-efficient communication medium in a Wireless-Body Area Network (WBAN). Wearable devices, communicating via radio frequency (RF)-based techniques incur higher losses and suffer from security vulnerabilities owing to their radiative nature. Since its emergence, Human Body Communication (HBC) has shown great promise by providing an energy-efficient and physically secure mode for information exchange among the wearable devices in the Electro-Quasistatics (EQS) frequency regime, which has been broadly explored. Besides, three regions in the body-channel characteristics namely EQS, Body Resonance (BR), and Device Resonance (DR), over a wide range of frequencies (100kHz-1GHz), have been conceptualized. With its high channel gain (CG) and bandwidth (BW), the BR-HBC can potentially exhilarate high data rate body-centric communications with wearable devices. However, the HBC channel in the BR regime that contributes to its high CG and BW demands an analysis of the influence of the body-tissue properties on the channel characteristics. This paper, for the first time, studies the effect of the electrical properties of the human body as a communication medium in the BR regime. Finite Element Method (FEM) based electromagnetic (EM) simulations and wearable-to-wearable (W-W) measurements are performed to study the channel variability with the electrical properties of the communication medium during its interaction with electromagnetic fields. Results are analyzed to develop an understanding of the EM-wave propagation in the vicinity of biological materials for biomedical applications.**

***Index Terms*—Human Body Communication (HBC), Body Resonance (BR), Finite Element Method (FEM), Wearable-to-Wearable (W-W) Measurements**

I. Introduction

Technology scaling in the last several decades has resulted in exponential growth in the usage of wearable devices around humans. However, many of these devices, especially those that require a lot of communication and processing power to support high data rate applications (like Audio, Video Streaming, AR-VR headsets, etc.), suffer from fairly short battery life. Now, optimizing the battery performance of these miniaturized devices requires the optimum selection of a communication medium and its corresponding mode for information exchange, a question that has drawn researcher's interest for many years. Wearable devices, when communicating through the air via RF-based techniques, owing to their radiative nature, incur a high loss and escalate the chances of eavesdropping from the surroundings. Owing to its ability to offer an energy-efficient and physically secure medium for communicating devices as a promising alternative to non-guided RF-based communication, the channel behavior for Human Body Communication in the Electro-Quasistatics (EQS) frequency regime has been widely studied. During the wide-band body-channel characterization for capacitive HBC [1], the Body Resonance frequency regime showed its potential to offer higher channel gain (CG) and wider bandwidth (BW) to support high data rate applications.

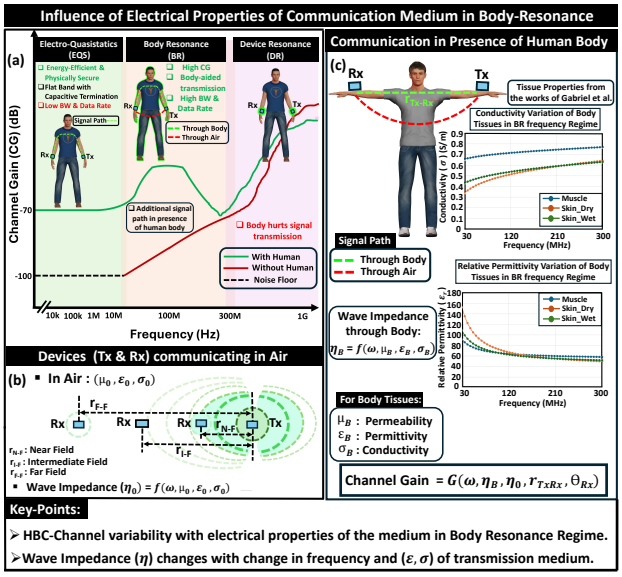


Fig. 1. Studying the effect of electrical properties of the body tissues that influence the propagation characteristics of capacitive Human Body Communication (HBC) using wearable devices (transmitter (Tx) and receiver (Rx)) in Body Resonance (BR) frequency regime: Schematic Diagram representing (a) Tx & Rx are communicating through air, (c) Human body with its lower wave impedance (η_B) than air, offers an additional path for signal propagation between the communicating devices.

Over the last several years, researchers have explored the body channel characteristics of humans while studying different modes of Intra and Inter-Body communication (IBC) [1]–[3]. While studying the propagation characteristics among different individuals by employing the human body as an antenna under different scenarios from 1 MHz to 90 MHz, Li et al. [4] investigated the influence of human tissues at 44 MHz (near the resonating frequency), to conclude that high-water-content tissues facilitate signal propagation. However, their use of a vector network analyzer, and baluns to isolate the ground electrodes for transmitter and receiver for measurements provided an optimistic estimation of the channel gain, which differs from the wearable-to-wearable

measurements. Introducing the relaxation time-based analysis to study the influence of the material properties of a communication structure on the channel variability in Human-Structure Interactions (HSI) in EQS, Sarkar et al. [5] proposed a circuit model to highlight the distributed nature of transmission medium. Nevertheless, no such study on the influence of electrical properties of body tissues for HBC in the BR regime exists in the literature that can facilitate unveiling the phenomenon that leads to the high CG and wide BW of the BR regime and hence intrigues an analysis of the properties of the communication medium, illustrated in Fig. 1.

II. Variation in Wave Impedance with Electrical Properties of Communication Medium

From definition, the electromagnetic wave impedance (η_M) in a medium-M can be defined as the impedance experienced by the EM wave when it travels through the medium. Now, the medium's electrical properties play a crucial role in deciding the wave impedance (η_M), and thereby possess a momentous influence on the wave propagation in the medium i.e.,

$$\eta_M = f(\omega, \mu_M, \sigma_M, \epsilon_M) = \sqrt{\left(\frac{j\omega\mu_M}{\sigma_M + j\omega\epsilon_M}\right)} = |\eta_M|\angle\theta_{\eta_M} \tag{1}$$

where μ_M, ϵ_M, and σ_M respectively represent magnetic permeability, the real part of electric permittivity, and electrical conductivity of the medium-M. Hence, The channel gain (CG) can be conceptually formulated as a function of the following parameters:

$$CG = G(\omega, \eta_M, r_{Tx-Rx}, \theta_{Rx}) \tag{2}$$

where r_{Tx-Rx} represents the distance between the Tx-Rx, and θ_{Rx} corresponds to the phase of the signal appearing at the Rx. In the BR frequency regime, when the devices (Tx and Rx) wirelessly communicate in the air, the propagating signal encounters the wave impedance of air. However, when a medium with different electrical conductivity and permittivity than air like the human body ($\sigma_{Body}(\omega)$, $\epsilon_{Body}(\omega)$) shown in Fig. 1 (c), is present in the vicinity of the devices, it offers an additional path for the signal propagation with its lower wave impedance, and influences the channel behavior.

III. Simulation Model & Results

A. Simulation Model

To study the effect of variation in electrical properties of a communication medium on HBC channel characteristics, Finite Element Methods (FEM) based EM simulations are carried out in the Body Resonance (BR) frequency regime (30 MHz to 250 MHz) using an EM solver-ANSYS High-Frequency Structure Simulator (HFSS). A cross-cylindrical model of the human body, emulating T-posture, shown in Fig. 2 (b), made up of body tissues (skin and muscle) [6] whose validity was confirmed [7] through comparison of EM field distribution with a complex human body model - VHP Female v2.2 from Neva Electromagnetics [8], is used to perform faster simulations at reduced complexity. The structural parameters of the body model and the communicating devices are illustrated in Fig. 2 ((a), (b), (c)).

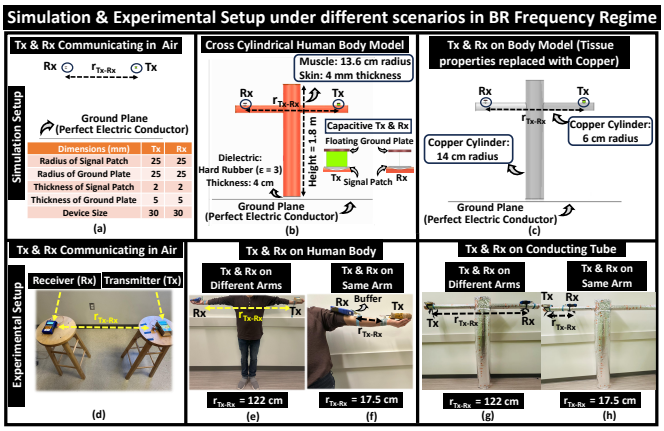


Fig. 2. Simulation Setup: Model used for FEM-based HFSS simulations and its structural parameters: (a) Without Body (i.e., Tx and Rx communicating in Air), (b) On-Body Tx & Rx, (c) Cross Cylindrical Copper Model. Experimental Setup: (d) Devices kept on wooden stools, Devices residing on subject's body: ((e) Opposite Arm and (f) Same Arm), Devices on Conducting Tube model: ((g) Opposite Arm and (h) Same Arm)

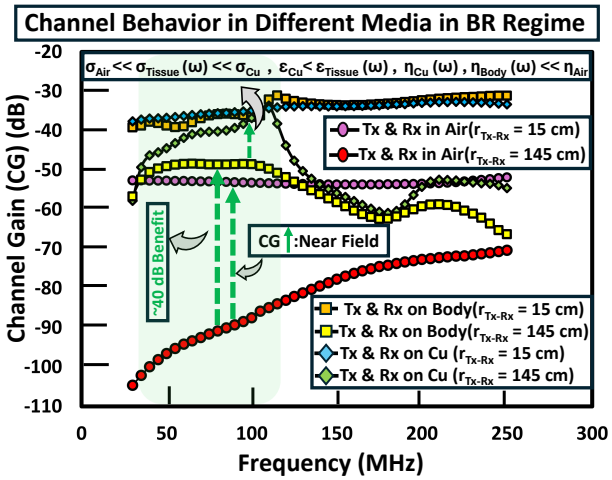


Fig. 3. Channel behavior in BR frequency regime (30 MHz to 250 MHz) with different communication mediums: Human Body and Copper, with their several orders of magnitude lower wave impedance (η_{Cu}, η_{Tissue} << η_{Air}) than air, offer higher channel gain (CG). The improvement in channel gain at reduced Tx-to-Rx distance (i.e., near field) is observable in all three scenarios.

B. Simulation Results

1) HBC Channel Behavior in the BR Frequency Regime: The variability in CG characteristics in the BR frequency regime in presence of different communication mediums (like air, human body, and copper) of varying electrical properties, are illustrated in Fig. 3. The variation in electrical properties (relative permittivity (ϵ_r) and conductivity (σ)) of the transmission medium leads to the variation in electromagnetic wave impedance (i.e., $\eta_{Cu} < \eta_{Body} << \eta_{Air}$). Owing to the wave impedance of air (η_{Air}~377Ω) which is orders of

magnitude higher than that of the human body and copper, the in-air device-to-device communication suffers from significantly higher (∼45 dB compared to on-body communication) path loss. Moreover, the CG difference between the copper model and the human body varies with operating frequency and primarily the phase (θ_{Rx}) of the received signal.

IV. Experimental Setup & Results

A. Experimental Setup

The complete measurement setup, comprising the subject's body posture, the communicating devices (Tx and Rx), and the measurement scenarios, are depicted in Fig. 2 ((d), (e,f), (g,h)). A handheld RF Explorer signal generator (RFE6GEN) is used to function as the transmitter, whose power level is kept at -5 dBm (∼0.5 V peak-to-peak) to keep the in-body field and current densities below the ICNIRP safety limits. A handheld RF Explorer spectrum analyzer (WSUB1G+) is employed to function as a wearable receiver (Rx). With its 50 Ω input impedance, the spectrum analyzer needs a buffer in series before it for high impedance capacitively terminated measurements. A protoboard is customized using BUF602, a high-speed (slew rate = 8000 V/μs) and wide bandwidth (1 GHz) buffer IC from Texas Instruments, and is powered up using a small-sized 3.7 V (nominal) rechargeable lipo battery. ESD wristbands are used to function as couplers.

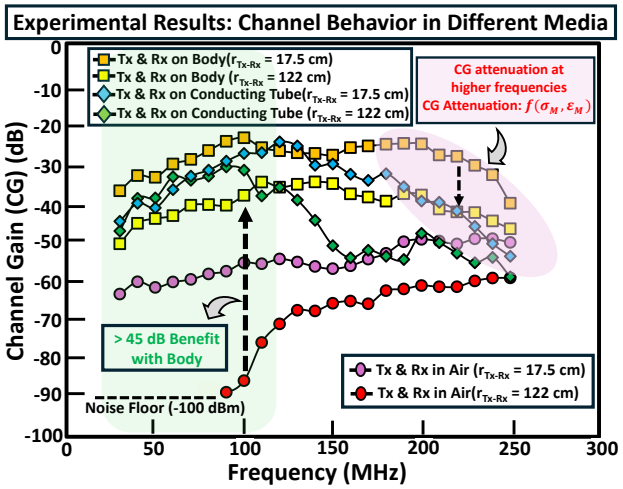


Fig. 4. Experimental Results: Medium with lower wave impedance (like Human Body and Conducting Tube) than air facilitates signal propagation from Tx to the Rx, and thereby improves the channel gain (CG). The CG benefits from the presence of a human body or other conducting medium start going down as the operating frequency approaches the RF range.

B. Experimental Results

The understandings of the proposed wave-impedance approach that are initially validated through numerical simulations are substantiated through measurements using wearable devices. The experiments were conducted by following the guidelines involving human subjects as approved by the Institutional Review Board (IRB), in a hall corridor to minimize the undesired couplings and interference from the surroundings. The measurement results for the variations in channel characteristics with operational frequency in the BR regime are presented in Fig. 4. The trends in experimental results are in good agreement with the simulation results. The obtained channel gain benefits (>∼45 dB) from the presence of the human body and conducting tube model, confirm the existence of an alternative signal path during information exchange between the Tx and Rx. However, with an increase in frequency towards RF, the benefit from the human body as a transmission medium goes down.

V. Conclusion

In conclusion, a study on the electrical properties of the transmission medium that decisively influences the channel characteristics for capacitive Human Body Communication in the Body Resonance frequency regime, is presented. Owing to the lower electromagnetic wave impedance of the body tissues (η_{Body}), the presence of the human body provides an appreciable benefit (>∼45 dB) in comparison to over-the-air device-to-device communication in the BR frequency regime. The effect of medium's electrical property becomes even more pronounced when the Rx is placed in the far field from the Tx. The wave propagation at the interface of the two mediums needs to be studied further to understand the phase of the received signal. This study can facilitate developing understanding of the propagation characteristics of the EM waves in the vicinity of body tissues that can be useful for enabling high data-rate body-centric communication channels.

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